

MIPT & YSDA · AUTUMN 2026

Deep Generative Models

Lecture 5

Roman Isachenko

Moscow Institute of Physics and Technology
Yandex School of Data Analysis

Recap of Previous Lecture

Theorem

$$\frac{1}{n} \sum_{i=1}^n D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}_i) \parallel p(\mathbf{z})) = D_{\text{KL}}(q_{\text{agg},\phi}(\mathbf{z}) \parallel p(\mathbf{z})) + \mathbb{I}_q[\mathbf{x}, \mathbf{z}].$$

Revisiting the ELBO

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n \mathcal{L}_{\phi,\theta}(\mathbf{x}_i) &= \underbrace{\frac{1}{n} \sum_{i=1}^n \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x}_i)} \log p_{\theta}(\mathbf{x}_i|\mathbf{z})}_{\text{Reconstruction Loss}} \\ &\quad - \underbrace{\mathbb{I}_q[\mathbf{x}, \mathbf{z}]}_{\text{Mutual Information}} - \underbrace{D_{\text{KL}}(q_{\text{agg},\phi}(\mathbf{z}) \parallel p(\mathbf{z}))}_{\text{Marginal KL}}. \end{aligned}$$

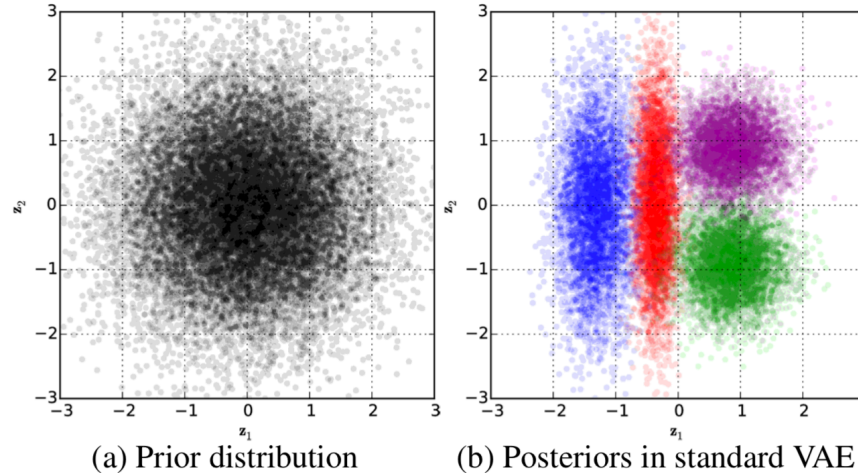
Optimal VAE Prior

$$D_{\text{KL}}(q_{\text{agg},\phi}(\mathbf{z}) \parallel p(\mathbf{z})) = 0 \Leftrightarrow p(\mathbf{z}) = q_{\text{agg},\phi}(\mathbf{z}) = \frac{1}{n} \sum_{i=1}^n q_{\phi}(\mathbf{z}|\mathbf{x}_i).$$

Thus, the optimal prior distribution $p(\mathbf{z})$ is the aggregated variational posterior $q_{\text{agg},\phi}(\mathbf{z})$.

Recap of Previous Lecture

- **Prior mismatch:** The unimodal encoder $q_\phi(\mathbf{z}|\mathbf{x})$ yields $q_{\text{agg},\phi}(\mathbf{z})$ that often does not match the Gaussian prior $p(\mathbf{z})$.
- **Blurriness from averaging:** With a fixed encoder, prior and Gaussian decoder variance, ELBO maximization gives $\mu^*(\mathbf{z}) = \mathbb{E}_{q_\phi(\mathbf{x}|\mathbf{z})}[\mathbf{x}]$. If distinct inputs $\mathbf{x} \neq \mathbf{x}'$ map to *overlapping* latent regions, the decoder averages over unrelated data.



Recap of Previous Lecture

Assumptions

- Let $c \sim \text{Cat}(\boldsymbol{\pi})$, where

$$\boldsymbol{\pi} = (\pi_1, \dots, \pi_K), \quad \pi_k = P(c = k), \quad \sum_{k=1}^K \pi_k = 1.$$

- For tokenizer training, fix $p(c) = \text{Uniform}\{1, \dots, K\}$.

ELBO

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_{q_{\phi}(c|\mathbf{x})} \log p_{\theta}(\mathbf{x}|c) - D_{\text{KL}}(q_{\phi}(c|\mathbf{x}) \parallel p(c)) \rightarrow \max_{\phi, \theta}.$$

$$D_{\text{KL}}(q_{\phi}(c|\mathbf{x}) \parallel p(c)) = -\text{H}(q_{\phi}(c|\mathbf{x})) + \log K.$$

Quantized Representation

Define the codebook (dictionary) space $\{\mathbf{e}_k\}_{k=1}^K$ with $\mathbf{e}_k \in \mathbb{R}^L$ and K the number of codebook entries.

$$\mathbf{z}_q = \mathbf{q}(\mathbf{z}) = \mathbf{e}_{k^*}, \quad \text{where } k^* = \arg \min_k \|\mathbf{z} - \mathbf{e}_k\|.$$

Recap of Previous Lecture

Quantize each of the $M = WH$ encoder vectors using the **same codebook**; the code map is $\mathbf{c} = (c_1, \dots, c_M)$, with $\mathbf{z}_{q,m} = \mathbf{e}_{c_m}$.

Deterministic Variational Posterior

$$q_{\phi}(c_m = k | \mathbf{x}) = \begin{cases} 1, & \text{for } k = \arg \min_j \|\mathbf{z}_{e,m} - \mathbf{e}_j\|; \\ 0, & \text{otherwise.} \end{cases}$$

ELBO

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_{q_{\phi}(\mathbf{c} | \mathbf{x})} \log p_{\theta}(\mathbf{x} | \mathbf{z}_q(\mathbf{c})) - M \log K = \log p_{\theta}(\mathbf{x} | \mathbf{z}_q) - M \log K.$$

Straight-Through Gradient Estimator

$$\frac{\partial \log p(\mathbf{x} | \mathbf{z}_q, \theta)}{\partial \phi} = \frac{\partial \log p_{\theta}(\mathbf{x} | \mathbf{z}_q)}{\partial \mathbf{z}_q} \cdot \frac{\partial \mathbf{z}_q}{\partial \phi} \approx \frac{\partial \log p_{\theta}(\mathbf{x} | \mathbf{z}_q)}{\partial \mathbf{z}_q} \cdot \frac{\partial \mathbf{z}_e}{\partial \phi}$$

The independent uniform prior is used during tokenizer training. Then fix the tokenizer and decoder, and fit an autoregressive prior $p_{\psi}(\mathbf{c})$ to the aggregated posterior $q_{\text{agg}, \phi}(\mathbf{c})$.

Recap of Previous Lecture

Likelihood-Free Learning

- Likelihood isn't always a suitable metric for evaluating generative models.
- Sometimes, the likelihood function can't even be computed exactly.

Imagine we have two sets of samples:

- $\{\mathbf{x}_i\}_{i=1}^{n_1} \sim p_{\text{data}}(\mathbf{x})$ — real samples;
- $\{\mathbf{x}_i\}_{i=1}^{n_2} \sim p_{\theta}(\mathbf{x})$ — generated (fake) samples.

$$p(y = 1|\mathbf{x}) = P(\mathbf{x} \sim p_{\text{data}}(\mathbf{x})); \quad p(y = 0|\mathbf{x}) = P(\mathbf{x} \sim p_{\theta}(\mathbf{x}))$$

Assumption

The generative model $p_{\theta}(\mathbf{x})$ matches $p_{\text{data}}(\mathbf{x})$ if a discriminative model $p(y|\mathbf{x})$ can't distinguish between them — that is, if $p(y = 1|\mathbf{x}) = 0.5$ for every $\mathbf{x}$.

- **Generator:** a generative model $\mathbf{x} = \mathbf{G}_{\theta}(\mathbf{z})$ that produces more realistic samples.
- **Discriminator:** a classifier $D_{\phi}(\mathbf{x}) \in [0, 1]$ distinguishing real from generated samples.

Recap of Previous Lecture

Cross-Entropy for Discriminative Model

$$\max_{p(y|\mathbf{x})} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log p(y = 1|\mathbf{x}) + \mathbb{E}_{p_{\theta}(\mathbf{x})} \log p(y = 0|\mathbf{x}) \right]$$

Recap of Previous Lecture

Cross-Entropy for Discriminative Model

$$\max_{p(y|\mathbf{x})} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log p(y = 1|\mathbf{x}) + \mathbb{E}_{p_{\theta}(\mathbf{x})} \log p(y = 0|\mathbf{x}) \right]$$

- **Discriminator:** A classifier $p_{\phi}(y = 1|\mathbf{x}) = D_{\phi}(\mathbf{x}) \in [0, 1]$, distinguishing real and generated samples. The discriminator aims to **maximize** cross-entropy.
- **Generator:** The generative model $\mathbf{x} = \mathbf{G}_{\theta}(\mathbf{z})$, $\mathbf{z} \sim p(\mathbf{z})$, seeks to fool the discriminator. The generator aims to **minimize** cross-entropy.

Recap of Previous Lecture

Cross-Entropy for Discriminative Model

$$\max_{p(y|\mathbf{x})} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log p(y = 1|\mathbf{x}) + \mathbb{E}_{p_{\theta}(\mathbf{x})} \log p(y = 0|\mathbf{x}) \right]$$

- **Discriminator:** A classifier $p_{\phi}(y = 1|\mathbf{x}) = D_{\phi}(\mathbf{x}) \in [0, 1]$, distinguishing real and generated samples. The discriminator aims to **maximize** cross-entropy.
- **Generator:** The generative model $\mathbf{x} = \mathbf{G}_{\theta}(\mathbf{z})$, $\mathbf{z} \sim p(\mathbf{z})$, seeks to fool the discriminator. The generator aims to **minimize** cross-entropy.

GAN Objective

$$\min_G \max_D \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p_{\theta}(\mathbf{x})} \log(1 - D(\mathbf{x})) \right]$$

Recap of Previous Lecture

Cross-Entropy for Discriminative Model

$$\max_{p(y|\mathbf{x})} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log p(y = 1|\mathbf{x}) + \mathbb{E}_{p_{\theta}(\mathbf{x})} \log p(y = 0|\mathbf{x}) \right]$$

- **Discriminator:** A classifier $p_{\phi}(y = 1|\mathbf{x}) = D_{\phi}(\mathbf{x}) \in [0, 1]$, distinguishing real and generated samples. The discriminator aims to **maximize** cross-entropy.
- **Generator:** The generative model $\mathbf{x} = \mathbf{G}_{\theta}(\mathbf{z})$, $\mathbf{z} \sim p(\mathbf{z})$, seeks to fool the discriminator. The generator aims to **minimize** cross-entropy.

GAN Objective

$$\min_G \max_D \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p_{\theta}(\mathbf{x})} \log(1 - D(\mathbf{x})) \right]$$

$$\min_G \max_D \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D(\mathbf{G}(\mathbf{z}))) \right]$$

Recap of Previous Lecture

Theorem

The following minimax game

$$\min_G \max_D \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D(\mathbf{G}(\mathbf{z}))) \right]$$

achieves its global optimum precisely when $p_{\text{data}}(\mathbf{x}) = p_{\theta}(\mathbf{x})$, and $D^*(\mathbf{x}) = 0.5$.

Expectations

If the generator can express **any** function and the discriminator is **optimal** at every step, the generator **will converge** to the target distribution.

Recap of Previous Lecture

Theorem

The following minimax game

$$\min_G \max_D \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D(\mathbf{G}(\mathbf{z}))) \right]$$

achieves its global optimum precisely when $p_{\text{data}}(\mathbf{x}) = p_{\theta}(\mathbf{x})$, and $D^*(\mathbf{x}) = 0.5$.

Expectations

If the generator can express **any** function and the discriminator is **optimal** at every step, the generator **will converge** to the target distribution.

Reality

- Generator updates are performed in parameter space, and the discriminator is often imperfectly optimized.
- Generator and discriminator losses typically oscillate during GAN training.

Outline

01 Generative Adversarial Networks (GAN)

02 Wasserstein Distance

03 Wasserstein GAN (WGAN)

04 Evaluation of Likelihood-Free Models

Frechet Inception Distance (FID) · Precision-Recall · CLIP Score · Human Eval

05 Langevin Dynamics

Outline

01 Generative Adversarial Networks (GAN)

02 Wasserstein Distance

03 Wasserstein GAN (WGAN)

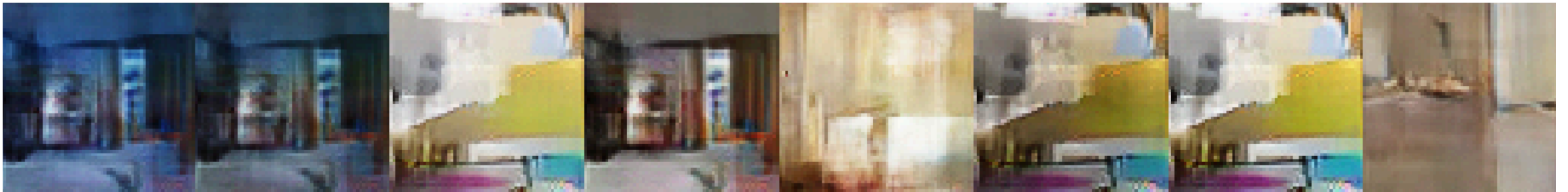
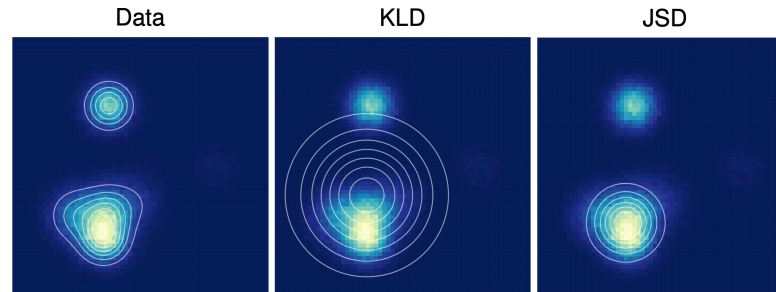
04 Evaluation of Likelihood-Free Models

Frechet Inception Distance (FID) · Precision-Recall · CLIP Score · Human Eval

05 Langevin Dynamics

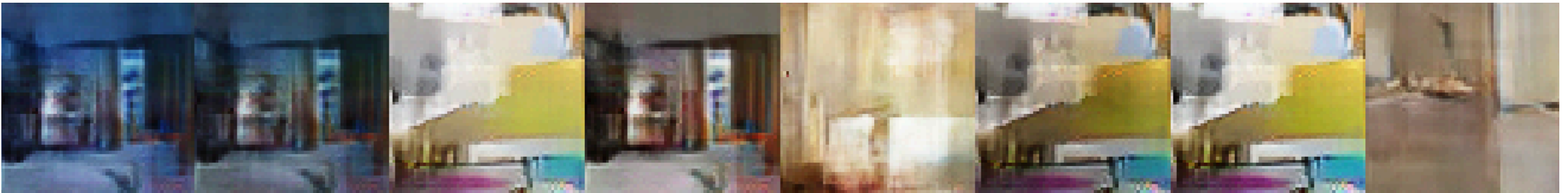
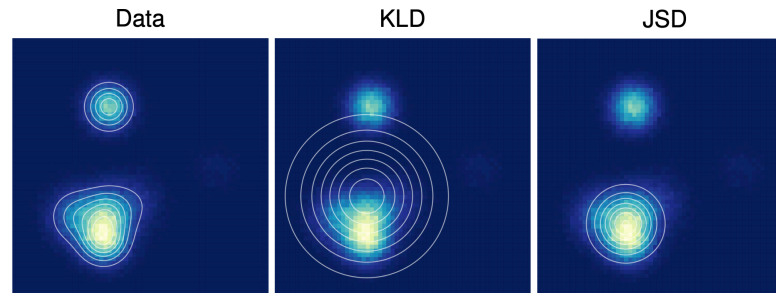
Mode Collapse

Mode collapse refers to the phenomenon where the generator in a GAN produces only one or a few different modes of the distribution.



Mode Collapse

Mode collapse refers to the phenomenon where the generator in a GAN produces only one or a few different modes of the distribution.



Numerous methods have been proposed to tackle mode collapse: changing architectures, adding regularization terms, injecting noise.

Jensen-Shannon vs Kullback-Leibler Divergences

- $p_{\text{data}}(\mathbf{x})$ is a fixed mixture of two Gaussians.
- $p(\mathbf{x}|\mu, \sigma) = \mathcal{N}(\mu, \sigma^2)$.

Mode Covering vs. Mode Seeking

$$D_{\text{KL}}(\pi \parallel p) = \int \pi(\mathbf{x}) \log \frac{\pi(\mathbf{x})}{p(\mathbf{x})} d\mathbf{x}, \quad D_{\text{KL}}(p \parallel \pi) = \int p(\mathbf{x}) \log \frac{p(\mathbf{x})}{\pi(\mathbf{x})} d\mathbf{x}$$
$$D_{\text{JS}}(\pi \parallel p) = \frac{1}{2} \left[D_{\text{KL}}\left(\pi(\mathbf{x}) \parallel \frac{\pi(\mathbf{x}) + p(\mathbf{x})}{2}\right) + D_{\text{KL}}\left(p(\mathbf{x}) \parallel \frac{\pi(\mathbf{x}) + p(\mathbf{x})}{2}\right) \right]$$

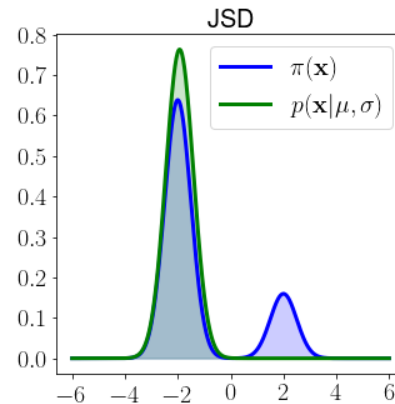
Jensen-Shannon vs Kullback-Leibler Divergences

- $p_{\text{data}}(\mathbf{x})$ is a fixed mixture of two Gaussians.
- $p(\mathbf{x}|\mu, \sigma) = \mathcal{N}(\mu, \sigma^2)$.

Mode Covering vs. Mode Seeking

$$D_{\text{KL}}(\pi \parallel p) = \int \pi(\mathbf{x}) \log \frac{\pi(\mathbf{x})}{p(\mathbf{x})} d\mathbf{x}, \quad D_{\text{KL}}(p \parallel \pi) = \int p(\mathbf{x}) \log \frac{p(\mathbf{x})}{\pi(\mathbf{x})} d\mathbf{x}$$

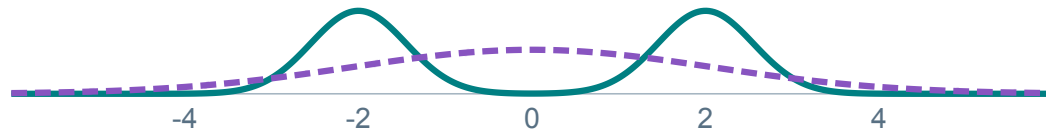
$$D_{\text{JS}}(\pi \parallel p) = \frac{1}{2} \left[D_{\text{KL}}\left(\pi(\mathbf{x}) \parallel \frac{\pi(\mathbf{x}) + p(\mathbf{x})}{2}\right) + D_{\text{KL}}\left(p(\mathbf{x}) \parallel \frac{\pi(\mathbf{x}) + p(\mathbf{x})}{2}\right) \right]$$



Mode Covering vs. Mode Seeking

$$p_{\text{data}}(x) = \frac{1}{2}\mathcal{N}(x \mid -2, 0.55^2) + \frac{1}{2}\mathcal{N}(x \mid 2, 0.55^2), \quad p_{\theta}(x) = \mathcal{N}(x \mid \mu, \sigma^2).$$

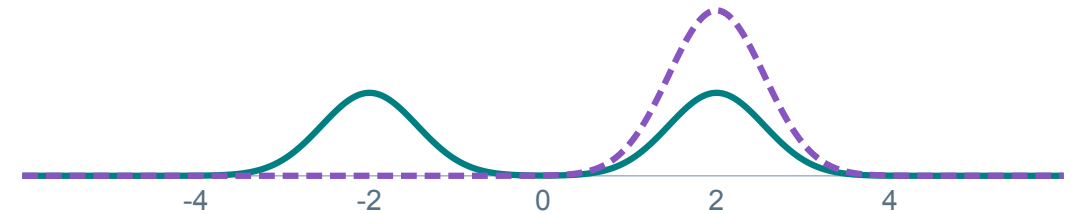
Forward KL optimum



$\mu = 0.00, \sigma = 2.07$

Forward KL 0.635 · Reverse KL 1.623

Reverse KL minimum (right mode)



$\mu = 2.00, \sigma = 0.55$

Forward KL 12.445 · Reverse KL 0.693

— Target: two modes

- - - - - Model: one Gaussian

Forward KL

Missing a target mode is expensive.

Reverse KL

Putting model mass in the low-density gap is expensive.

This difference comes from restricting the model to one Gaussian. If it could match the target exactly, both KLs would be zero.

From VQ-VAE to VQGAN

Same **encoder** → **quantization** → **decoder**; a new reconstruction objective.



Same spatial downsampling $f = 8$ and codebook size $K = 8192$.

From VQ-VAE to VQGAN

Same **encoder** → **quantization** → **decoder**; a new reconstruction objective.



Same spatial downsampling $f = 8$ and codebook size $K = 8192$.

Perceptual Loss

A **fixed pretrained network** F extracts features:

$$\mathcal{L}_{\text{perc}} = \|F(\mathbf{x}) - F(\hat{\mathbf{x}})\|_2^2.$$

Adversarial Loss

Train discriminator D to separate real and reconstructed **image patches**.

Train the autoencoder to **fool** D .

Generative Modeling in Latent Space

First learn a compact representation; then learn to **generate its latent codes**.

Discrete: VQGAN

Quantization gives a map of code indices.

$$\mathbf{c} \in \{1, \dots, K\}^{W \times H}$$

Continuous: KL-Autoencoder

Perceptual + adversarial losses; weak KL.

$$\mathbf{z} \in \mathbb{R}^{W \times H \times L}$$

Generative Modeling in Latent Space

First learn a compact representation; then learn to **generate its latent codes**.

Discrete: VQGAN

Quantization gives a map of code indices.

$$\mathbf{c} \in \{1, \dots, K\}^{W \times H}$$

Autoregressive prior over the code map.

$$\mathbf{c} \sim p_{\psi}(\mathbf{c}) \longrightarrow \mathbf{z}_q(\mathbf{c}) \longrightarrow \hat{\mathbf{x}}$$

Sample codes $\rightarrow$ lookup $\rightarrow$ decode.

Continuous: KL-Autoencoder

Perceptual + adversarial losses; weak KL.

$$\mathbf{z} \in \mathbb{R}^{W \times H \times L}$$

Diffusion prior in the learned latent space.

$$\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \longrightarrow \mathbf{z} \sim p_{\psi}(\mathbf{z}) \longrightarrow \hat{\mathbf{x}}$$

Denoise in latent space $\rightarrow$ decode.

Generative Modeling in Latent Space

First learn a compact representation; then learn to **generate its latent codes**.

Discrete: VQGAN

Quantization gives a map of code indices.

$$\mathbf{c} \in \{1, \dots, K\}^{W \times H}$$

Autoregressive prior over the code map.

$$\mathbf{c} \sim p_\psi(\mathbf{c}) \longrightarrow \mathbf{z}_q(\mathbf{c}) \longrightarrow \hat{\mathbf{x}}$$

Sample codes $\rightarrow$ lookup $\rightarrow$ decode.

Continuous: KL-Autoencoder

Perceptual + adversarial losses; weak KL.

$$\mathbf{z} \in \mathbb{R}^{W \times H \times L}$$

Diffusion prior in the learned latent space.

$$\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \longrightarrow \mathbf{z} \sim p_\psi(\mathbf{z}) \longrightarrow \hat{\mathbf{x}}$$

Denoise in latent space $\rightarrow$ decode.

Both recipes freeze the autoencoder before learning a prior that models its **aggregated posterior**.

Outline

01 Generative Adversarial Networks (GAN)

02 Wasserstein Distance

03 Wasserstein GAN (WGAN)

04 Evaluation of Likelihood-Free Models

Frechet Inception Distance (FID) · Precision-Recall · CLIP Score · Human Eval

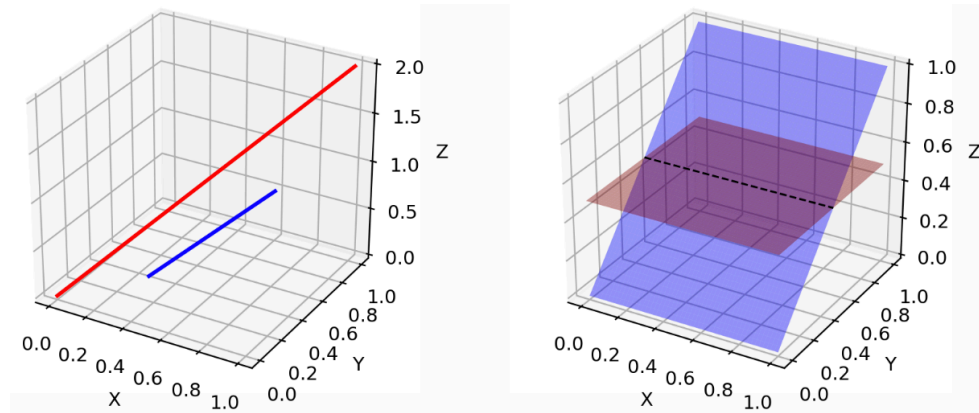
05 Langevin Dynamics

Theoretical Results

- The dimensionality of $\mathbf{z}$ is less than that of $\mathbf{x}$, so $p_\theta(\mathbf{x})$ with $\mathbf{x} = \mathbf{G}_\theta(\mathbf{z})$ lives on a low-dimensional manifold.

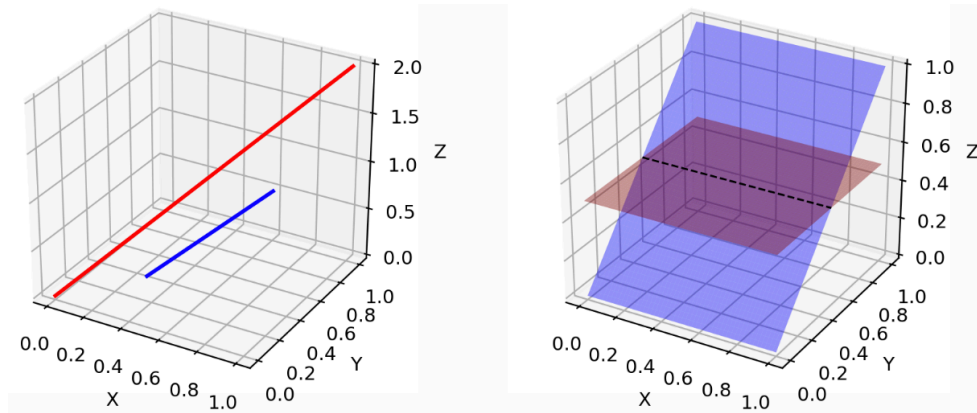
Theoretical Results

- The dimensionality of $\mathbf{z}$ is less than that of $\mathbf{x}$, so $p_{\theta}(\mathbf{x})$ with $\mathbf{x} = \mathbf{G}_{\theta}(\mathbf{z})$ lives on a low-dimensional manifold.
- The true data distribution $p_{\text{data}}(\mathbf{x})$ is also supported on a low-dimensional manifold.



Theoretical Results

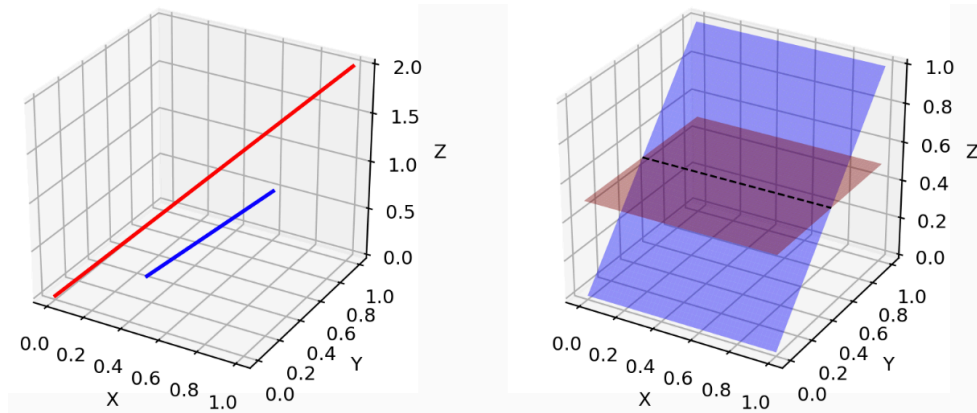
- The dimensionality of $\mathbf{z}$ is less than that of $\mathbf{x}$, so $p_{\theta}(\mathbf{x})$ with $\mathbf{x} = \mathbf{G}_{\theta}(\mathbf{z})$ lives on a low-dimensional manifold.
- The true data distribution $p_{\text{data}}(\mathbf{x})$ is also supported on a low-dimensional manifold.



- If $p_{\text{data}}(\mathbf{x})$ and $p_{\theta}(\mathbf{x})$ are disjoint, a smooth optimal discriminator can exist!

Theoretical Results

- The dimensionality of $\mathbf{z}$ is less than that of $\mathbf{x}$, so $p_{\theta}(\mathbf{x})$ with $\mathbf{x} = \mathbf{G}_{\theta}(\mathbf{z})$ lives on a low-dimensional manifold.
- The true data distribution $p_{\text{data}}(\mathbf{x})$ is also supported on a low-dimensional manifold.



- If $p_{\text{data}}(\mathbf{x})$ and $p_{\theta}(\mathbf{x})$ are disjoint, a smooth optimal discriminator can exist!

For such low-dimensional, disjoint manifolds:

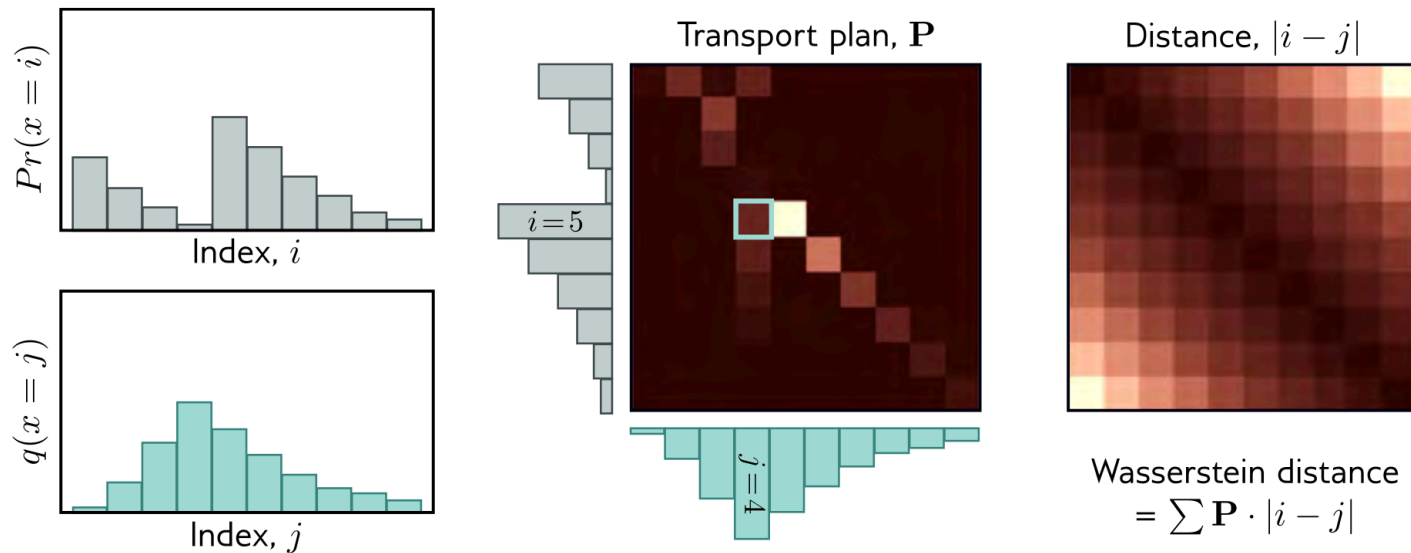
$$D_{\text{KL}}(p_{\text{data}} \parallel p_{\theta}) = D_{\text{KL}}(p_{\theta} \parallel p_{\text{data}}) = \infty, \quad D_{\text{JS}}(p_{\text{data}} \parallel p_{\theta}) = \log 2$$

Wasserstein Distance (Discrete)

Also known as the **Earth Mover's Distance**.

Optimal Transport Formulation

The minimum cost of moving and transforming a pile of “dirt” shaped like one probability distribution to match another.



Wasserstein Distance (Continuous)

$$\begin{aligned} W(\pi \| p) &= \inf_{\gamma \in \Gamma(\pi, p)} \mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\| \\ &= \inf_{\gamma \in \Gamma(\pi, p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\| \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 d\mathbf{x}_2 \end{aligned}$$

- $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ is the transport plan: the amount of “dirt” assigned from $\mathbf{x}_1$ to $\mathbf{x}_2$.

$$\int \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 = p(\mathbf{x}_2); \quad \int \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_2 = \pi(\mathbf{x}_1).$$

- $\Gamma(\pi, p)$ denotes the set of all joint distributions $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ with marginals π and p .
- $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ is the mass, $\|\mathbf{x}_1 - \mathbf{x}_2\|$ is the distance.

Wasserstein Distance (Continuous)

$$\begin{aligned} W(\pi \| p) &= \inf_{\gamma \in \Gamma(\pi, p)} \mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\| \\ &= \inf_{\gamma \in \Gamma(\pi, p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\| \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 d\mathbf{x}_2 \end{aligned}$$

- $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ is the transport plan: the amount of “dirt” assigned from $\mathbf{x}_1$ to $\mathbf{x}_2$.

$$\int \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 = p(\mathbf{x}_2); \quad \int \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_2 = \pi(\mathbf{x}_1).$$

- $\Gamma(\pi, p)$ denotes the set of all joint distributions $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ with marginals π and p .
- $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ is the mass, $\|\mathbf{x}_1 - \mathbf{x}_2\|$ is the distance.

Wasserstein Metric

$$W_s(\pi, p) = \inf_{\gamma \in \Gamma(\pi, p)} \left(\mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\|^s \right)^{1/s}$$

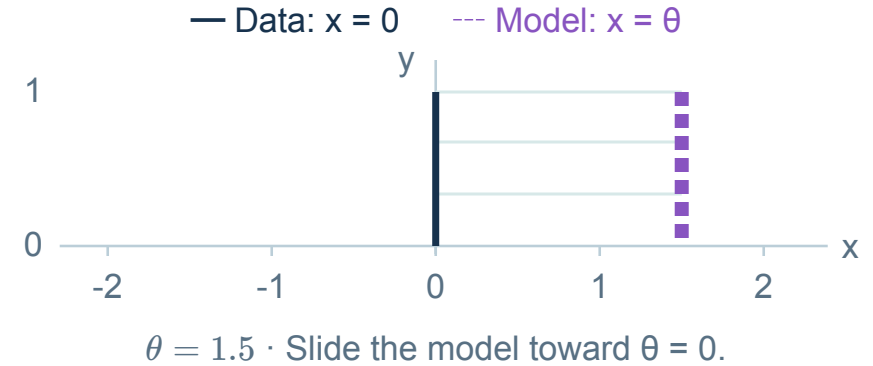
In our setting, $W(\pi \| p) = W_1(\pi, p)$, which is the transport cost formulation.

In Lecture 12, minibatch optimal transport pairs noise and data samples for flow matching.

Wasserstein Distance vs KL vs JSD

Consider two-dimensional distributions:

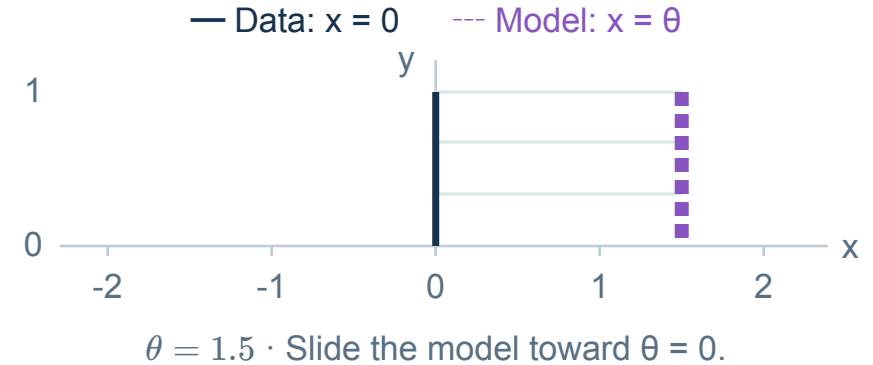
$$p_{\text{data}}(x, y) = (0, U[0, 1]), \quad p_{\theta}(x, y) = (\theta, U[0, 1])$$



Wasserstein Distance vs KL vs JSD

Consider two-dimensional distributions:

$$p_{\text{data}}(x, y) = (0, U[0, 1]), \quad p_{\theta}(x, y) = (\theta, U[0, 1])$$



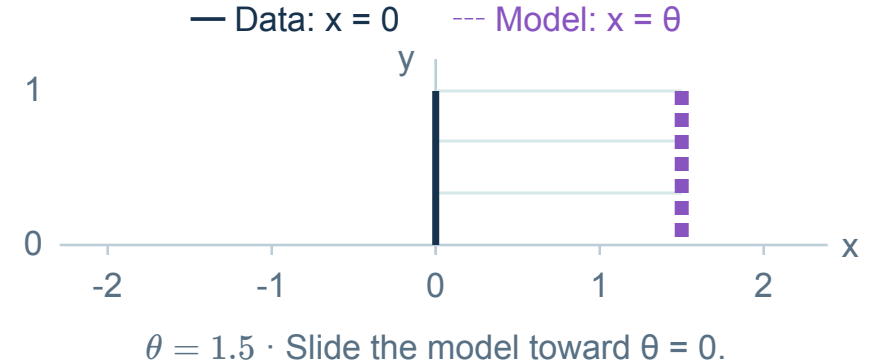
$\theta = 0$: Both distributions are identical.

$$D_{\text{KL}}(p_{\text{data}} \| p_{\theta}) = D_{\text{KL}}(p_{\theta} \| p_{\text{data}}) = D_{\text{JS}}(p_{\theta} \| p_{\text{data}}) = W(p_{\text{data}} \| p_{\theta}) = 0$$

Wasserstein Distance vs KL vs JSD

Consider two-dimensional distributions:

$$p_{\text{data}}(x, y) = (0, U[0, 1]), \quad p_{\theta}(x, y) = (\theta, U[0, 1])$$



$\theta = 0$: Both distributions are identical.

$$D_{\text{KL}}(p_{\text{data}} \| p_{\theta}) = D_{\text{KL}}(p_{\theta} \| p_{\text{data}}) = D_{\text{JS}}(p_{\theta} \| p_{\text{data}}) = W(p_{\text{data}} \| p_{\theta}) = 0$$

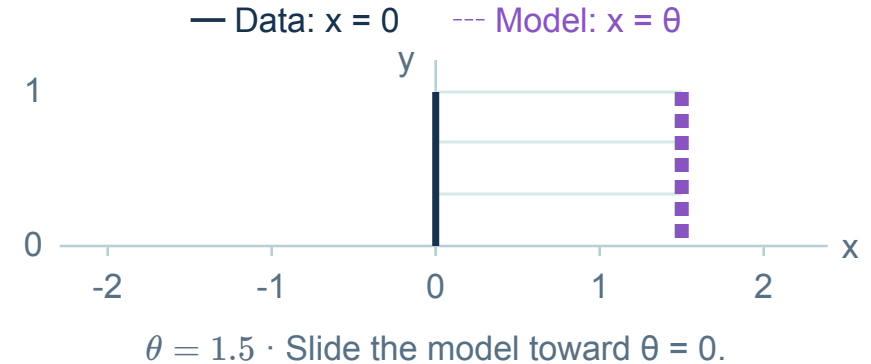
$\theta \neq 0$:

$$D_{\text{KL}}(p_{\text{data}} \| p_{\theta}) = \int_{U[0,1]} 1 \log \frac{1}{0} dy = \infty = D_{\text{KL}}(p_{\theta} \| p_{\text{data}})$$

Wasserstein Distance vs KL vs JSD

Consider two-dimensional distributions:

$$p_{\text{data}}(x, y) = (0, U[0, 1]), \quad p_{\theta}(x, y) = (\theta, U[0, 1])$$



$\theta = 0$: Both distributions are identical.

$$D_{\text{KL}}(p_{\text{data}} \| p_{\theta}) = D_{\text{KL}}(p_{\theta} \| p_{\text{data}}) = D_{\text{JS}}(p_{\theta} \| p_{\text{data}}) = W(p_{\text{data}} \| p_{\theta}) = 0$$

$\theta \neq 0$:

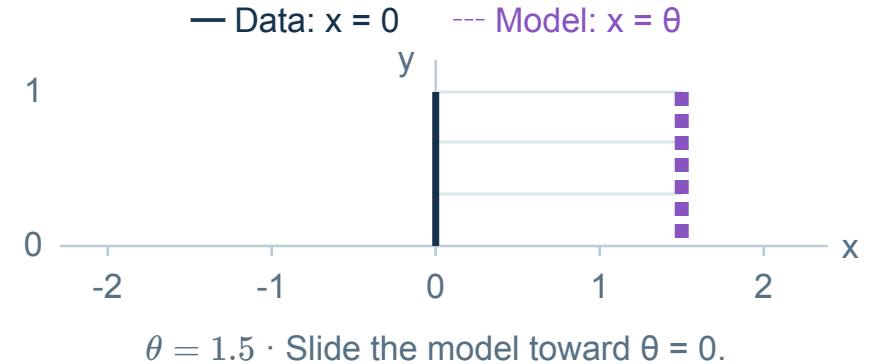
$$D_{\text{KL}}(p_{\text{data}} \| p_{\theta}) = \int_{U[0,1]} 1 \log \frac{1}{0} dy = \infty = D_{\text{KL}}(p_{\theta} \| p_{\text{data}})$$

$$D_{\text{JS}}(p_{\text{data}} \| p_{\theta}) = \frac{1}{2} \left(\int_{U[0,1]} 1 \log \frac{1}{1/2} dy + \int_{U[0,1]} 1 \log \frac{1}{1/2} dy \right) = \log 2$$

Wasserstein Distance vs KL vs JSD

Consider two-dimensional distributions:

$$p_{\text{data}}(x, y) = (0, U[0, 1]), \quad p_{\theta}(x, y) = (\theta, U[0, 1])$$



$\theta = 0$: Both distributions are identical.

$$D_{\text{KL}}(p_{\text{data}} \| p_{\theta}) = D_{\text{KL}}(p_{\theta} \| p_{\text{data}}) = D_{\text{JS}}(p_{\theta} \| p_{\text{data}}) = W(p_{\text{data}} \| p_{\theta}) = 0$$

$\theta \neq 0$:

$$D_{\text{KL}}(p_{\text{data}} \| p_{\theta}) = \int_{U[0,1]} 1 \log \frac{1}{0} dy = \infty = D_{\text{KL}}(p_{\theta} \| p_{\text{data}})$$

$$D_{\text{JS}}(p_{\text{data}} \| p_{\theta}) = \frac{1}{2} \left(\int_{U[0,1]} 1 \log \frac{1}{1/2} dy + \int_{U[0,1]} 1 \log \frac{1}{1/2} dy \right) = \log 2$$

$$W(p_{\text{data}} \| p_{\theta}) = |\theta|$$

Why Wasserstein Distance?

On a compact data space $\mathcal{X}$:

Dependence on Generator Parameters

For a feedforward generator G_θ with Lipschitz activations and $\mathbb{E}_{p(\mathbf{z})} \|\mathbf{z}\| < \infty$, $W(p_{\text{data}} \| p_\theta)$ is continuous in θ and differentiable almost everywhere.

Why Wasserstein Distance?

On a compact data space $\mathcal{X}$:

Dependence on Generator Parameters

For a feedforward generator G_θ with Lipschitz activations and $\mathbb{E}_{p(\mathbf{z})} \|\mathbf{z}\| < \infty$, $W(p_{\text{data}} \| p_\theta)$ is continuous in θ and differentiable almost everywhere.

Convergence of Distributions

For a fixed distribution π and a sequence p_t on $\mathcal{X}$, as $t \rightarrow \infty$:

$$D_{\text{KL}}(\pi \| p_t) \rightarrow 0 \implies D_{\text{JS}}(\pi \| p_t) \rightarrow 0 \implies W(\pi \| p_t) \rightarrow 0$$

The first implication also holds for $D_{\text{KL}}(p_t \| \pi) \rightarrow 0$; the converses need not hold.

Wasserstein can capture distributions getting closer even when their supports do not overlap.

Outline

01 Generative Adversarial Networks (GAN)

02 Wasserstein Distance

03 Wasserstein GAN (WGAN)

04 Evaluation of Likelihood-Free Models

Frechet Inception Distance (FID) · Precision-Recall · CLIP Score · Human Eval

05 Langevin Dynamics

Wasserstein GAN

Wasserstein Distance

$$\begin{aligned} W(\pi \| p) &= \inf_{\gamma \in \Gamma(\pi, p)} \mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\| \\ &= \inf_{\gamma \in \Gamma(\pi, p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\| \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 d\mathbf{x}_2 \end{aligned}$$

Wasserstein GAN

Wasserstein Distance

$$\begin{aligned} W(\pi||p) &= \inf_{\gamma \in \Gamma(\pi, p)} \mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\| \\ &= \inf_{\gamma \in \Gamma(\pi, p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\| \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 d\mathbf{x}_2 \end{aligned}$$

The infimum over all possible $\gamma \in \Gamma(\pi, p)$ is computationally intractable.

Wasserstein GAN

Wasserstein Distance

$$\begin{aligned} W(\pi \| p) &= \inf_{\gamma \in \Gamma(\pi, p)} \mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\| \\ &= \inf_{\gamma \in \Gamma(\pi, p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\| \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 d\mathbf{x}_2 \end{aligned}$$

The infimum over all possible $\gamma \in \Gamma(\pi, p)$ is computationally intractable.

Theorem (Kantorovich-Rubinstein Duality)

$$W(\pi \| p) = \frac{1}{K} \max_{\|f\|_L \leq K} \left[\mathbb{E}_{\pi(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p(\mathbf{x})} f(\mathbf{x}) \right]$$

where $f : \mathbb{R}^m \rightarrow \mathbb{R}$ is K -Lipschitz ($\|f\|_L \leq K$):

$$|f(\mathbf{x}_1) - f(\mathbf{x}_2)| \leq K \|\mathbf{x}_1 - \mathbf{x}_2\|, \quad \forall \mathbf{x}_1, \mathbf{x}_2 \in \mathcal{X}.$$

Wasserstein GAN

Wasserstein Distance

$$\begin{aligned} W(\pi \| p) &= \inf_{\gamma \in \Gamma(\pi, p)} \mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\| \\ &= \inf_{\gamma \in \Gamma(\pi, p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\| \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 d\mathbf{x}_2 \end{aligned}$$

The infimum over all possible $\gamma \in \Gamma(\pi, p)$ is computationally intractable.

Theorem (Kantorovich-Rubinstein Duality)

$$W(\pi \| p) = \frac{1}{K} \max_{\|f\|_L \leq K} \left[\mathbb{E}_{\pi(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p(\mathbf{x})} f(\mathbf{x}) \right]$$

where $f : \mathbb{R}^m \rightarrow \mathbb{R}$ is K -Lipschitz ($\|f\|_L \leq K$):

$$|f(\mathbf{x}_1) - f(\mathbf{x}_2)| \leq K \|\mathbf{x}_1 - \mathbf{x}_2\|, \quad \forall \mathbf{x}_1, \mathbf{x}_2 \in \mathcal{X}.$$

We can thus estimate $W(\pi \| p)$ using only samples and a function f .

Wasserstein GAN

Theorem (Kantorovich-Rubinstein Duality)

$$W(p_{\text{data}} \| p_{\theta}) = \frac{1}{K} \max_{\|f\|_L \leq K} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p_{\theta}(\mathbf{x})} f(\mathbf{x}) \right]$$

- We must ensure that f is K -Lipschitz continuous.
- Let $f_{\phi}(\mathbf{x})$ be a feedforward neural network parameterized by ϕ .
- If the weights ϕ are restricted to a compact set Φ , then f_{ϕ} is K -Lipschitz.

Wasserstein GAN

Theorem (Kantorovich-Rubinstein Duality)

$$W(p_{\text{data}} \| p_{\theta}) = \frac{1}{K} \max_{\|f\|_L \leq K} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p_{\theta}(\mathbf{x})} f(\mathbf{x}) \right]$$

- We must ensure that f is K -Lipschitz continuous.
- Let $f_{\phi}(\mathbf{x})$ be a feedforward neural network parameterized by ϕ .
- If the weights ϕ are restricted to a compact set Φ , then f_{ϕ} is K -Lipschitz.
- Clamp weights within the box $\Phi = [-c, c]^d$ (e.g. $c = 0.01$) after each update.

$$\begin{aligned} K \cdot W(p_{\text{data}} \| p_{\theta}) &= \max_{\|f\|_L \leq K} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p_{\theta}(\mathbf{x})} f(\mathbf{x}) \right] \\ &\geq \max_{\phi \in \Phi} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f_{\phi}(\mathbf{x}) - \mathbb{E}_{p_{\theta}(\mathbf{x})} f_{\phi}(\mathbf{x}) \right] \end{aligned}$$

Wasserstein GAN

Standard GAN Objective

$$\min_{\theta} \max_{\phi} \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D_{\phi}(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D_{\phi}(\mathbf{G}_{\theta}(\mathbf{z})))$$

WGAN Objective

$$\min_{\theta} W(p_{\text{data}} || p_{\theta}) \approx \min_{\theta} \max_{\phi \in \Phi} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f_{\phi}(\mathbf{x}) - \mathbb{E}_{p(\mathbf{z})} f_{\phi}(\mathbf{G}_{\theta}(\mathbf{z})) \right]$$

Wasserstein GAN

Standard GAN Objective

$$\min_{\theta} \max_{\phi} \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D_{\phi}(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D_{\phi}(\mathbf{G}_{\theta}(\mathbf{z})))$$

WGAN Objective

$$\min_{\theta} W(p_{\text{data}} || p_{\theta}) \approx \min_{\theta} \max_{\phi \in \Phi} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f_{\phi}(\mathbf{x}) - \mathbb{E}_{p(\mathbf{z})} f_{\phi}(\mathbf{G}_{\theta}(\mathbf{z})) \right]$$

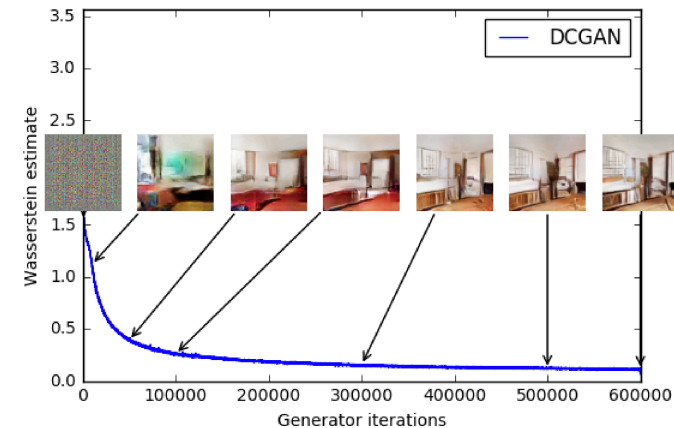
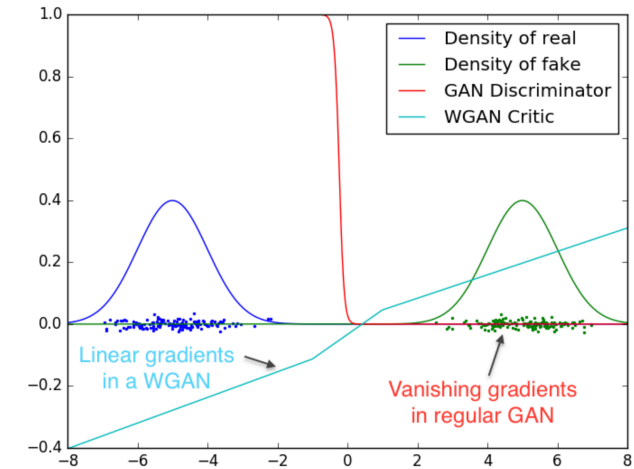
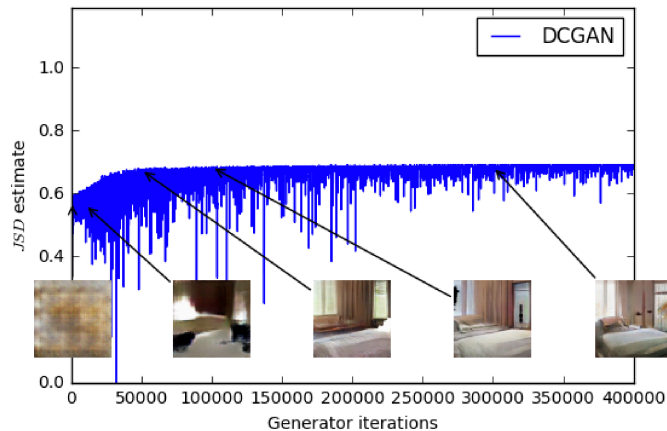
- The discriminator D is replaced by function f : in WGAN, it is known as the **critic**, which is *not* a classifier.
- “*Weight clipping is a clearly terrible way to enforce a Lipschitz constraint.*”
 - If c is large, optimizing the critic is hard; if c is small, gradients may vanish.

Gradient penalty encourages the critic’s input-gradient norm to stay close to 1 on interpolations between real and generated samples.

Spectral normalization controls layer-wise Lipschitz constants by dividing weight matrices by their largest singular values.

Wasserstein GAN

- WGAN provides nonzero gradients even if distributions' supports are disjoint.
- $D_{JS}(p_{\text{data}} || p_{\theta})$ is poorly correlated with sample quality and remains near its maximum value $\log 2 \approx 0.69$.
- $W(p_{\text{data}} || p_{\theta})$ is tightly correlated with quality.



Outline

01 Generative Adversarial Networks (GAN)

02 Wasserstein Distance

03 Wasserstein GAN (WGAN)

04 Evaluation of Likelihood-Free Models

Frechet Inception Distance (FID) · Precision-Recall · CLIP Score · Human Eval

05 Langevin Dynamics

What Makes a Good Generator?

Likelihood alone does not tell us whether samples are useful for the target task.

Illustrative failure cases created for this lecture.

Theis L. et al. A Note on the Evaluation of Generative Models, 2016 · Kynkäänniemi T. et al. Improved Precision and Recall Metric for Assessing Generative Models, 2019

Lin Z. et al. Evaluating Text-to-Visual Generation with Image-to-Text Generation, 2024

What Makes a Good Generator?

Likelihood alone does not tell us whether samples are useful for the target task.

Fidelity

Does each sample look plausible?



Sharp, but anatomically wrong.

What Makes a Good Generator?

Likelihood alone does not tell us whether samples are useful for the target task.

Fidelity

Does each sample look plausible?



Sharp, but anatomically wrong.

Coverage & Diversity

Does the set cover the data?



The same output on every run.

What Makes a Good Generator?

Likelihood alone does not tell us whether samples are useful for the target task.

Fidelity

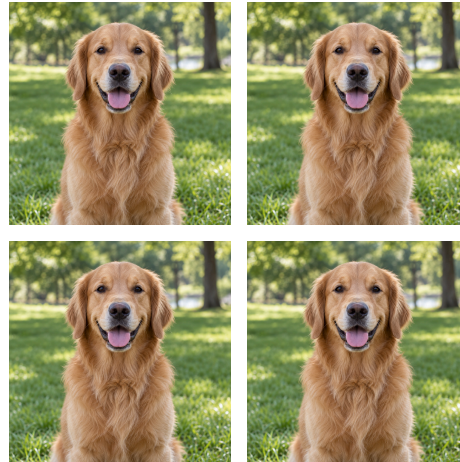
Does each sample look plausible?



Sharp, but anatomically wrong.

Coverage & Diversity

Does the set cover the data?



The same output on every run.

Condition Alignment

Does it follow the condition?



Prompt: “Three red apples.”
Result: only two.

Outline

01 Generative Adversarial Networks (GAN)

02 Wasserstein Distance

03 Wasserstein GAN (WGAN)

04 Evaluation of Likelihood-Free Models

Frechet Inception Distance (FID) · Precision-Recall · CLIP Score · Human Eval

05 Langevin Dynamics

Wasserstein Metric

$$W_s(\pi||p) = \inf_{\gamma \in \Gamma(\pi, p)} \left(\mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\|^s \right)^{1/s}$$

Wasserstein Metric

$$W_s(\pi||p) = \inf_{\gamma \in \Gamma(\pi, p)} \left(\mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\|^s \right)^{1/s}$$

Wasserstein GAN (Optimal Transport)

$$\begin{aligned} W(\pi||p) &= \inf_{\gamma \in \Gamma(\pi, p)} \mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\| \\ &= \inf_{\gamma \in \Gamma(\pi, p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\| \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 d\mathbf{x}_2 \end{aligned}$$

Wasserstein Metric

$$W_s(\pi||p) = \inf_{\gamma \in \Gamma(\pi, p)} \left(\mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\|^s \right)^{1/s}$$

Wasserstein GAN (Optimal Transport)

$$\begin{aligned} W(\pi||p) &= \inf_{\gamma \in \Gamma(\pi, p)} \mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\| \\ &= \inf_{\gamma \in \Gamma(\pi, p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\| \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 d\mathbf{x}_2 \end{aligned}$$

Theorem

If $\pi(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_\pi, \boldsymbol{\Sigma}_\pi)$, $p(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_p, \boldsymbol{\Sigma}_p)$, then

$$W_2^2(\pi||p) = \|\boldsymbol{\mu}_\pi - \boldsymbol{\mu}_p\|^2 + \text{tr} \left[\boldsymbol{\Sigma}_\pi + \boldsymbol{\Sigma}_p - 2 \left(\boldsymbol{\Sigma}_\pi^{1/2} \boldsymbol{\Sigma}_p \boldsymbol{\Sigma}_\pi^{1/2} \right)^{1/2} \right]$$

Wasserstein Metric

$$W_s(\pi||p) = \inf_{\gamma \in \Gamma(\pi, p)} \left(\mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\|^s \right)^{1/s}$$

Wasserstein GAN (Optimal Transport)

$$\begin{aligned} W(\pi||p) &= \inf_{\gamma \in \Gamma(\pi, p)} \mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\| \\ &= \inf_{\gamma \in \Gamma(\pi, p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\| \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 d\mathbf{x}_2 \end{aligned}$$

Theorem

If $\pi(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_\pi, \boldsymbol{\Sigma}_\pi)$, $p(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_p, \boldsymbol{\Sigma}_p)$, then

$$W_2^2(\pi||p) = \|\boldsymbol{\mu}_\pi - \boldsymbol{\mu}_p\|^2 + \text{tr} \left[\boldsymbol{\Sigma}_\pi + \boldsymbol{\Sigma}_p - 2 \left(\boldsymbol{\Sigma}_\pi^{1/2} \boldsymbol{\Sigma}_p \boldsymbol{\Sigma}_\pi^{1/2} \right)^{1/2} \right]$$

Frechet Inception Distance

$$\text{FID}(p_{\text{data}}, p_{\boldsymbol{\theta}}) = W_2^2(p_{\text{data}}||p_{\boldsymbol{\theta}})$$

Frechet Inception Distance (FID)

$$\text{FID}(p_{\text{data}}, p_{\theta}) = \|\boldsymbol{\mu}_{\text{data}} - \boldsymbol{\mu}_{\theta}\|^2 + \text{tr} \left[\boldsymbol{\Sigma}_{\text{data}} + \boldsymbol{\Sigma}_{\theta} - 2 \left(\boldsymbol{\Sigma}_{\text{data}}^{1/2} \boldsymbol{\Sigma}_{\theta} \boldsymbol{\Sigma}_{\text{data}}^{1/2} \right)^{1/2} \right]$$

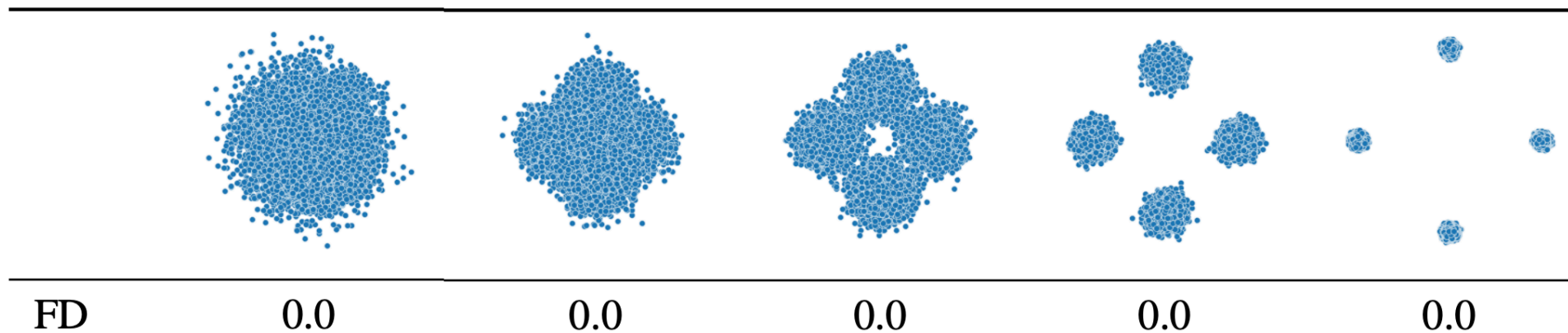
- FID uses Inception image embeddings $\mathbf{z} = \mathbf{f}(\mathbf{x})$.
- $\boldsymbol{\mu}_{\text{data}}$, $\boldsymbol{\Sigma}_{\text{data}}$ and $\boldsymbol{\mu}_{\theta}$, $\boldsymbol{\Sigma}_{\theta}$ are statistics of latent representations for samples from $p_{\text{data}}(\mathbf{x})$ and $p_{\theta}(\mathbf{x})$.
- **FD-DINOv2** uses the same formula with DINOv2 features; the Gaussian approximation remains.

Frechet Inception Distance (FID)

$$\text{FID}(p_{\text{data}}, p_{\theta}) = \|\mu_{\text{data}} - \mu_{\theta}\|^2 + \text{tr} \left[\Sigma_{\text{data}} + \Sigma_{\theta} - 2 \left(\Sigma_{\text{data}}^{1/2} \Sigma_{\theta} \Sigma_{\text{data}}^{1/2} \right)^{1/2} \right]$$

- FID uses Inception image embeddings $\mathbf{z} = \mathbf{f}(\mathbf{x})$.
- μ_{data} , Σ_{data} and μ_{θ} , Σ_{θ} are statistics of latent representations for samples from $p_{\text{data}}(\mathbf{x})$ and $p_{\theta}(\mathbf{x})$.
- **FD-DINOv2** uses the same formula with DINOv2 features; the Gaussian approximation remains.

$\text{FID}(p(\mathbf{x}), \mathcal{N}(0, \mathbf{I}))$



Frechet Inception Distance (FID)

$$\text{FID}(p_{\text{data}}, p_{\theta}) = \|\boldsymbol{\mu}_{\text{data}} - \boldsymbol{\mu}_{\theta}\|^2 + \text{tr} \left[\boldsymbol{\Sigma}_{\text{data}} + \boldsymbol{\Sigma}_{\theta} - 2 \left(\boldsymbol{\Sigma}_{\text{data}}^{1/2} \boldsymbol{\Sigma}_{\theta} \boldsymbol{\Sigma}_{\text{data}}^{1/2} \right)^{1/2} \right]$$

Frechet Inception Distance (FID)

$$\text{FID}(p_{\text{data}}, p_{\theta}) = \|\mu_{\text{data}} - \mu_{\theta}\|^2 + \text{tr} \left[\Sigma_{\text{data}} + \Sigma_{\theta} - 2 \left(\Sigma_{\text{data}}^{1/2} \Sigma_{\theta} \Sigma_{\text{data}}^{1/2} \right)^{1/2} \right]$$

Drawbacks

- Depends on the pretrained classification network.
- Uses the normality assumption.
- May not correlate with human evaluation.

Finite-sample FID is biased, and its value depends on the sample size.

Model	Model-A	Model-B
FID	21.40	18.42
FID _∞	20.16	17.19
Human rater preference	92.5%	6.9%

Maximum Mean Discrepancy (MMD)

The earlier figure shows that matching means and covariances can miss changes in shape. For embedding distributions P, Q , compare means of richer nonlinear features ψ :

$$\boldsymbol{\mu}_P = \mathbb{E}_{\mathbf{z} \sim P} \psi(\mathbf{z}), \quad \boldsymbol{\mu}_Q = \mathbb{E}_{\mathbf{w} \sim Q} \psi(\mathbf{w}).$$

Maximum Mean Discrepancy (MMD)

The earlier figure shows that matching means and covariances can miss changes in shape. For embedding distributions P, Q , compare means of richer nonlinear features ψ :

$$\boldsymbol{\mu}_P = \mathbb{E}_{\mathbf{z} \sim P} \psi(\mathbf{z}), \quad \boldsymbol{\mu}_Q = \mathbb{E}_{\mathbf{w} \sim Q} \psi(\mathbf{w}).$$

Kernel Trick

$$k(\mathbf{a}, \mathbf{u}) = \langle \psi(\mathbf{a}), \psi(\mathbf{u}) \rangle.$$

Compute inner products directly, without explicitly constructing ψ .

Maximum Mean Discrepancy (MMD)

The earlier figure shows that matching means and covariances can miss changes in shape. For embedding distributions P, Q , compare means of richer nonlinear features ψ :

$$\boldsymbol{\mu}_P = \mathbb{E}_{\mathbf{z} \sim P} \psi(\mathbf{z}), \quad \boldsymbol{\mu}_Q = \mathbb{E}_{\mathbf{w} \sim Q} \psi(\mathbf{w}).$$

Kernel Trick

$$k(\mathbf{a}, \mathbf{u}) = \langle \psi(\mathbf{a}), \psi(\mathbf{u}) \rangle.$$

Compute inner products directly, without explicitly constructing ψ .

Distance Between Mean Features

Let $\mathbf{z}, \mathbf{z}' \sim P$ and $\mathbf{w}, \mathbf{w}' \sim Q$ be mutually independent.

$$\begin{aligned} \text{MMD}_k^2(P, Q) &= \|\boldsymbol{\mu}_P - \boldsymbol{\mu}_Q\|^2 \\ &= \langle \boldsymbol{\mu}_P, \boldsymbol{\mu}_P \rangle + \langle \boldsymbol{\mu}_Q, \boldsymbol{\mu}_Q \rangle - 2\langle \boldsymbol{\mu}_P, \boldsymbol{\mu}_Q \rangle \\ &= \underbrace{\mathbb{E}k(\mathbf{z}, \mathbf{z}')}_{\text{within } P} + \underbrace{\mathbb{E}k(\mathbf{w}, \mathbf{w}')}_{\text{within } Q} - 2 \underbrace{\mathbb{E}k(\mathbf{z}, \mathbf{w})}_{\text{between } P, Q}. \end{aligned}$$

CMMD: Comparing CLIP Image Embeddings

CMMD = squared MMD on **CLIP image embeddings**.

$$\text{MMD}_k^2(P, Q) = \mathbb{E}k(\mathbf{z}, \mathbf{z}') + \mathbb{E}k(\mathbf{w}, \mathbf{w}') - 2\mathbb{E}k(\mathbf{z}, \mathbf{w}).$$

$$k(\mathbf{z}, \mathbf{w}) = \exp\left(-\frac{\|\mathbf{z} - \mathbf{w}\|^2}{2\sigma^2}\right), \quad \sigma > 0.$$

One Gaussian kernel already corresponds to infinitely many nonlinear features in ψ .

$$\text{MMD}_k(P, Q) = 0 \iff P = Q.$$

CMMD: Comparing CLIP Image Embeddings

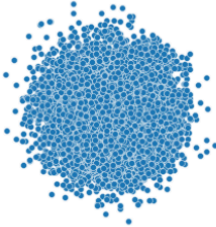
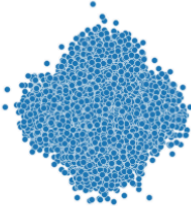
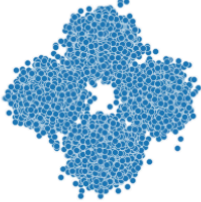
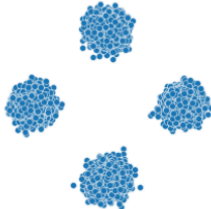
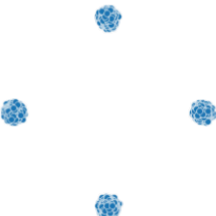
CMMD = squared MMD on **CLIP image embeddings**.

$$\text{MMD}_k^2(P, Q) = \mathbb{E}k(\mathbf{z}, \mathbf{z}') + \mathbb{E}k(\mathbf{w}, \mathbf{w}') - 2\mathbb{E}k(\mathbf{z}, \mathbf{w}).$$

$$k(\mathbf{z}, \mathbf{w}) = \exp\left(-\frac{\|\mathbf{z} - \mathbf{w}\|^2}{2\sigma^2}\right), \quad \sigma > 0.$$

One Gaussian kernel already corresponds to infinitely many nonlinear features in ψ .

$$\text{MMD}_k(P, Q) = 0 \iff P = Q.$$

					
FD	0.0	0.0	0.0	0.0	0.0
MMD	0.0	5.794	17.21	78.88	202.8

Synthetic example: same means and covariances, different shapes.

Outline

01 Generative Adversarial Networks (GAN)

02 Wasserstein Distance

03 Wasserstein GAN (WGAN)

04 Evaluation of Likelihood-Free Models

Frechet Inception Distance (FID) · **Precision-Recall** · CLIP Score · Human Eval

05 Langevin Dynamics

Precision-Recall

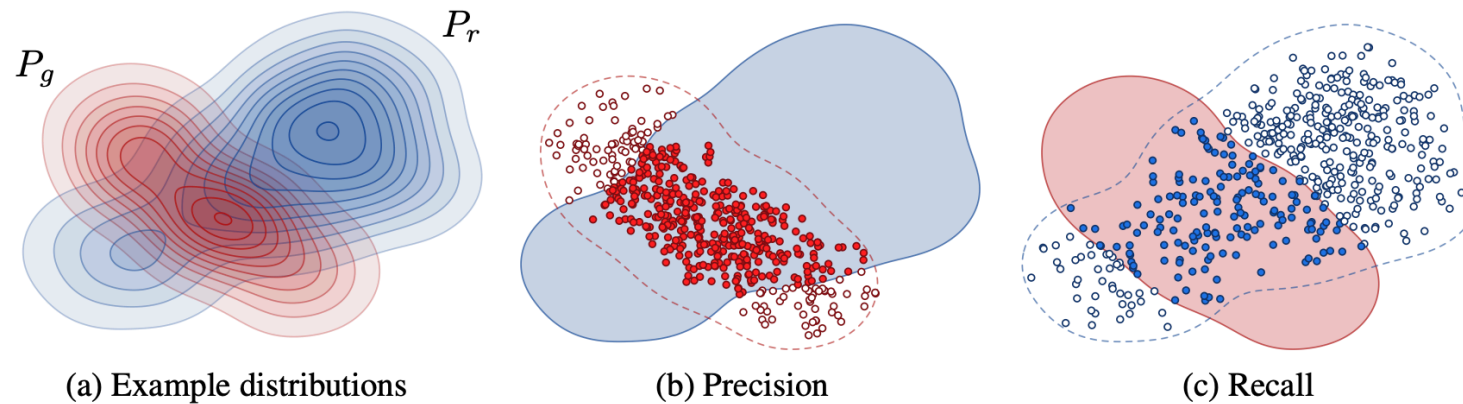
Desirable Properties for Samples

- **Sharpness:** generated samples should possess high visual quality.
- **Diversity:** their variation should match that in the training data.

Precision-Recall

Desirable Properties for Samples

- **Sharpness:** generated samples should possess high visual quality.
- **Diversity:** their variation should match that in the training data.



- **Precision** denotes the fraction of generated images that look realistic.
- **Recall** measures how well the generator covers the training data manifold.

Precision-Recall

- $\mathcal{S}_{\text{data}} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\text{data}}(\mathbf{x})$ — real samples;
- $\mathcal{S}_{\theta} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\theta}(\mathbf{x})$ — generated samples.

Precision-Recall

- $\mathcal{S}_{\text{data}} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\text{data}}(\mathbf{x})$ — real samples;
- $\mathcal{S}_{\theta} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\theta}(\mathbf{x})$ — generated samples.

Define a binary function:

$$\mathbb{I}(\mathbf{x}, \mathcal{S}) = \begin{cases} 1, & \text{if } \exists \mathbf{x}' \in \mathcal{S} : \|\mathbf{x} - \mathbf{x}'\|_2 \leq \|\mathbf{x}' - \text{NN}_k(\mathbf{x}', \mathcal{S})\|_2; \\ 0, & \text{otherwise.} \end{cases}$$

Precision-Recall

- $\mathcal{S}_{\text{data}} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\text{data}}(\mathbf{x})$ — real samples;
- $\mathcal{S}_{\theta} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\theta}(\mathbf{x})$ — generated samples.

Define a binary function:

$$\mathbb{I}(\mathbf{x}, \mathcal{S}) = \begin{cases} 1, & \text{if } \exists \mathbf{x}' \in \mathcal{S} : \|\mathbf{x} - \mathbf{x}'\|_2 \leq \|\mathbf{x}' - \text{NN}_k(\mathbf{x}', \mathcal{S})\|_2; \\ 0, & \text{otherwise.} \end{cases}$$

$$\text{Pr}(\mathcal{S}_{\text{data}}, \mathcal{S}_{\theta}) = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{S}_{\theta}} \mathbb{I}(\mathbf{x}, \mathcal{S}_{\text{data}}); \quad \text{Rec}(\mathcal{S}_{\text{data}}, \mathcal{S}_{\theta}) = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{S}_{\text{data}}} \mathbb{I}(\mathbf{x}, \mathcal{S}_{\theta}).$$

Precision-Recall

- $\mathcal{S}_{\text{data}} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\text{data}}(\mathbf{x})$ — real samples;
- $\mathcal{S}_{\theta} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\theta}(\mathbf{x})$ — generated samples.

Define a binary function:

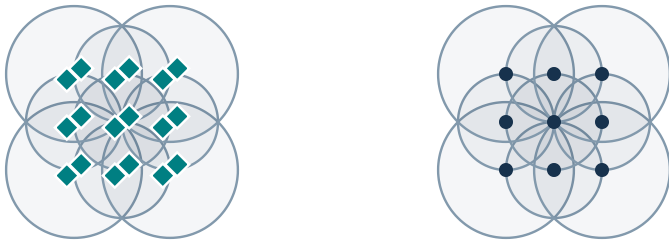
$$\mathbb{I}(\mathbf{x}, \mathcal{S}) = \begin{cases} 1, & \text{if } \exists \mathbf{x}' \in \mathcal{S} : \|\mathbf{x} - \mathbf{x}'\|_2 \leq \|\mathbf{x}' - \text{NN}_k(\mathbf{x}', \mathcal{S})\|_2; \\ 0, & \text{otherwise.} \end{cases}$$

$$\text{Pr}(\mathcal{S}_{\text{data}}, \mathcal{S}_{\theta}) = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{S}_{\theta}} \mathbb{I}(\mathbf{x}, \mathcal{S}_{\text{data}}); \quad \text{Rec}(\mathcal{S}_{\text{data}}, \mathcal{S}_{\theta}) = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{S}_{\text{data}}} \mathbb{I}(\mathbf{x}, \mathcal{S}_{\theta}).$$

Precision

Test generated ♦ in real balls

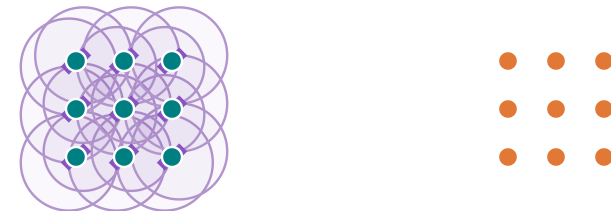
18 / 18



Recall

Test real • in generated balls

9 / 18



Precision-Recall

- $\mathcal{S}_{\text{data}} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\text{data}}(\mathbf{x})$ — real samples;
- $\mathcal{S}_{\theta} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\theta}(\mathbf{x})$ — generated samples.

Define a binary function:

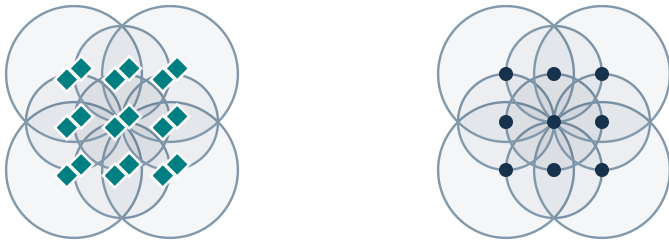
$$\mathbb{I}(\mathbf{x}, \mathcal{S}) = \begin{cases} 1, & \text{if } \exists \mathbf{x}' \in \mathcal{S} : \|\mathbf{x} - \mathbf{x}'\|_2 \leq \|\mathbf{x}' - \text{NN}_k(\mathbf{x}', \mathcal{S})\|_2; \\ 0, & \text{otherwise.} \end{cases}$$

$$\text{Pr}(\mathcal{S}_{\text{data}}, \mathcal{S}_{\theta}) = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{S}_{\theta}} \mathbb{I}(\mathbf{x}, \mathcal{S}_{\text{data}}); \quad \text{Rec}(\mathcal{S}_{\text{data}}, \mathcal{S}_{\theta}) = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{S}_{\text{data}}} \mathbb{I}(\mathbf{x}, \mathcal{S}_{\theta}).$$

Precision

Test generated ♦ in real balls

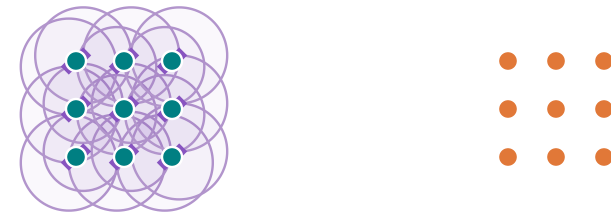
18 / 18



Recall

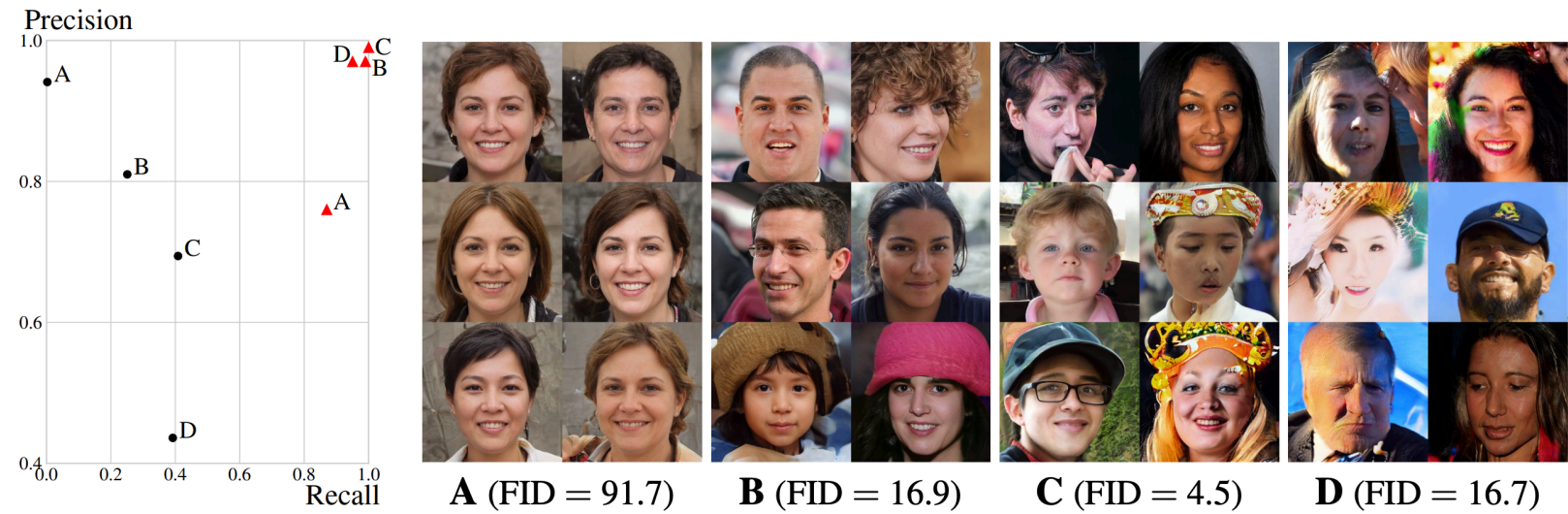
Test real • in generated balls

9 / 18

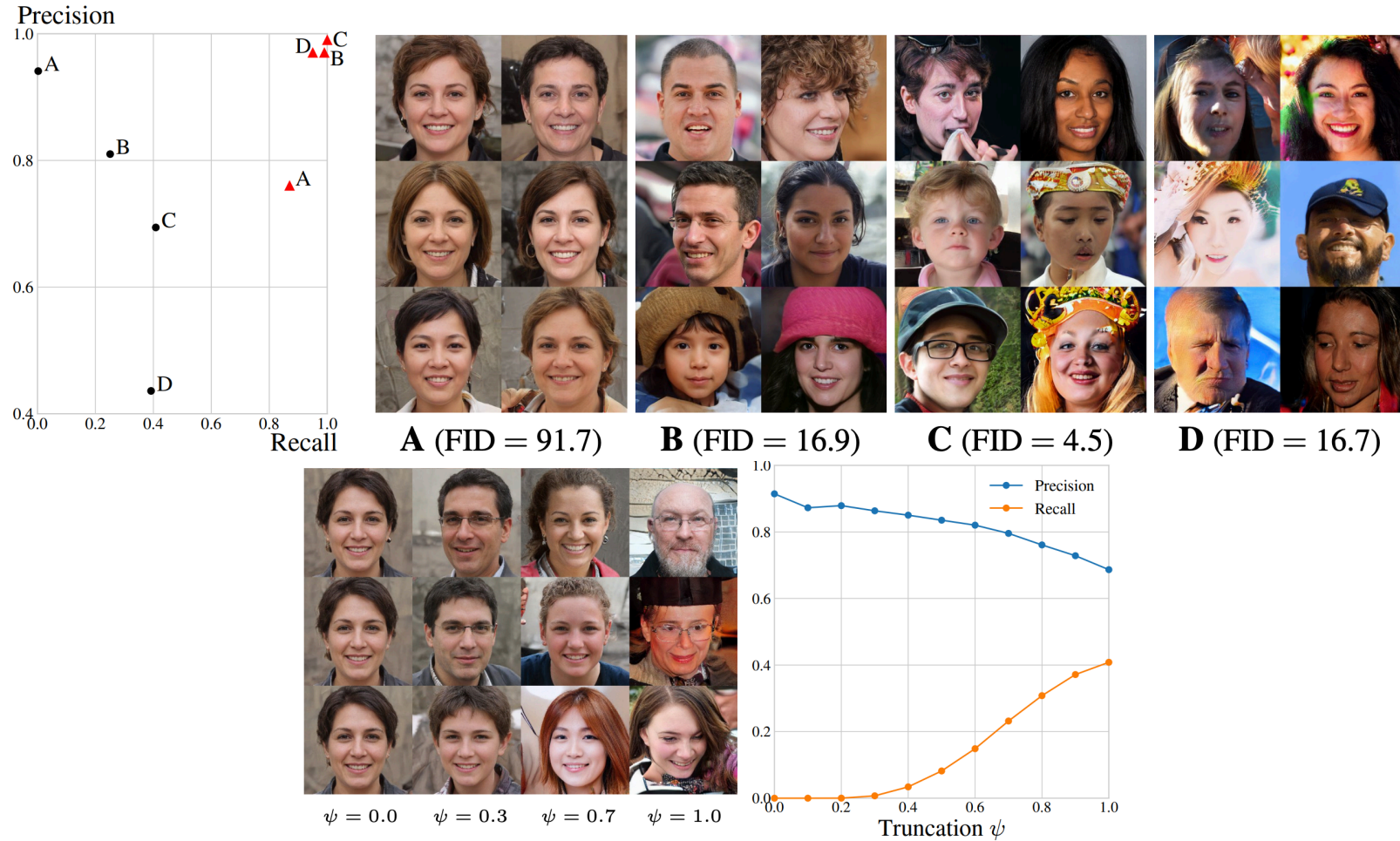


Embed the samples using a pretrained network (as in FID).

Precision-Recall



Precision-Recall



Here, stronger truncation (smaller ψ) favors precision over recall.

Outline

01 Generative Adversarial Networks (GAN)

02 Wasserstein Distance

03 Wasserstein GAN (WGAN)

04 Evaluation of Likelihood-Free Models

Frechet Inception Distance (FID) · Precision-Recall · **CLIP Score** · Human Eval

05 Langevin Dynamics

CLIP Similarity and Human Preferences

Score an image $\mathbf{x}$ for a text prompt t .

CLIP: Learn from Image–Caption Pairs

Map images and text into a shared embedding space:

$$s_{\text{CLIP}}(\mathbf{x}, t) = \cos(\mathbf{f}_{\text{img}}(\mathbf{x}), \mathbf{f}_{\text{text}}(t)) .$$

CLIP Similarity and Human Preferences

Score an image $\mathbf{x}$ for a text prompt t .

CLIP: Learn from Image–Caption Pairs

Map images and text into a shared embedding space:

$$s_{\text{CLIP}}(\mathbf{x}, t) = \cos(\mathbf{f}_{\text{img}}(\mathbf{x}), \mathbf{f}_{\text{text}}(t)).$$

PickScore: Learn from Human Choices

Fine-tune CLIP on pairs: for the same prompt, a human prefers $\mathbf{x}^+$ to $\mathbf{x}^-$.

$$\mathbb{P}_{\theta}(\mathbf{x}^+ \succ \mathbf{x}^- \mid t) = \frac{\exp r_{\theta}(\mathbf{x}^+, t)}{\exp r_{\theta}(\mathbf{x}^+, t) + \exp r_{\theta}(\mathbf{x}^-, t)}.$$

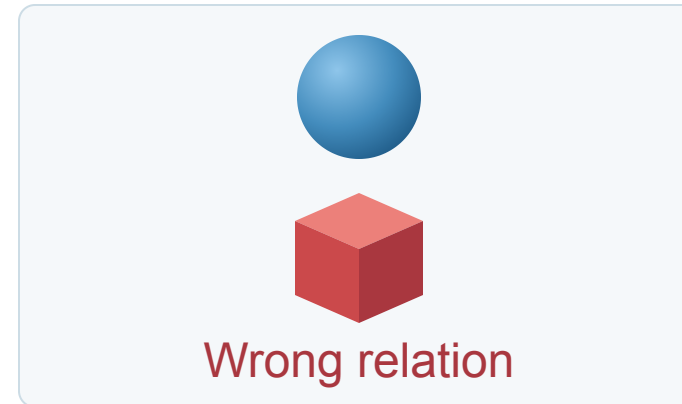
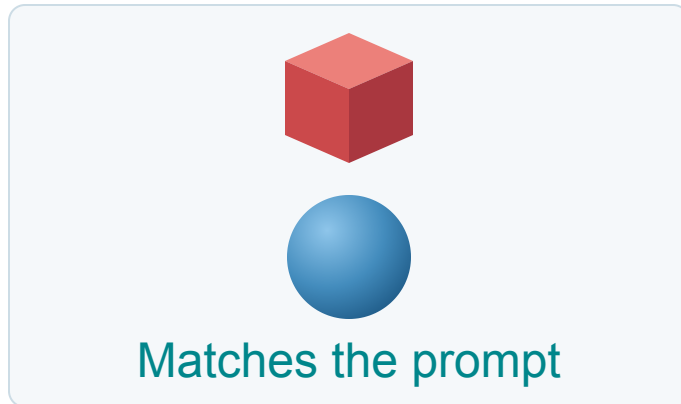
Train the score r_{θ} to assign higher probability to the human choice.

Other preference models: [ImageReward](#), [HPSv2](#).

A learned preference score can guide sample selection and model training — but optimizing the score can exploit its flaws.

Does the Image Follow the Prompt?

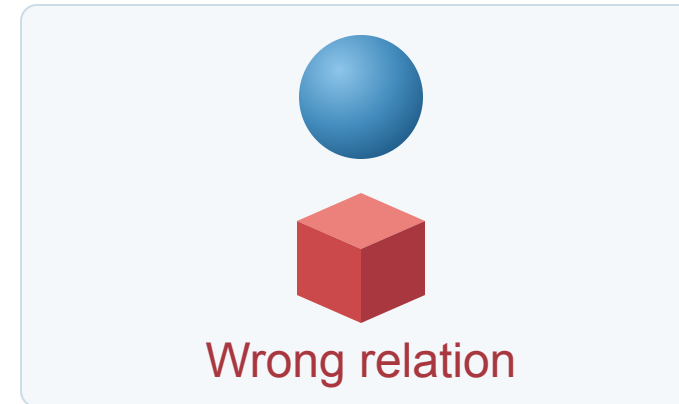
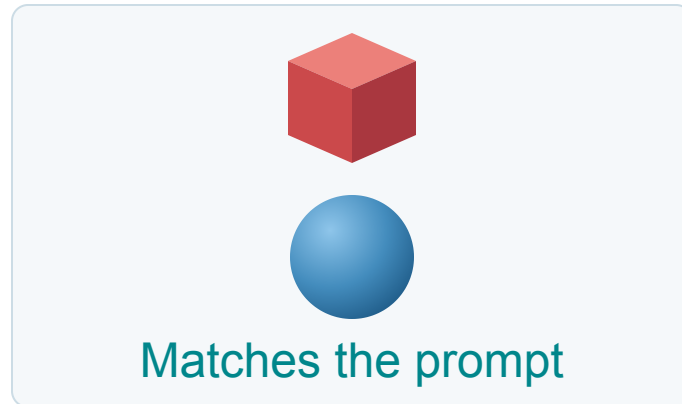
Prompt t : “A red cube above a blue sphere.”



Same objects and colors; opposite spatial relation. CLIP similarity can miss this.

Does the Image Follow the Prompt?

Prompt t : “A red cube above a blue sphere.”



Same objects and colors; opposite spatial relation. CLIP similarity can miss this.

VQAScore: Ask a Visual Question

$q(t)$: “Does this image show a red cube above a blue sphere?”

$$s_{\text{VQA}}(\mathbf{x}, t) = p_{\text{VQA}}(\text{Yes} \mid \mathbf{x}, q(t)).$$

GenEval / DPG-Bench: test objects, counts, attributes, and relations across prompts.

Outline

01 Generative Adversarial Networks (GAN)

02 Wasserstein Distance

03 Wasserstein GAN (WGAN)

04 Evaluation of Likelihood-Free Models

Frechet Inception Distance (FID) · Precision-Recall · CLIP Score · **Human Eval**

05 Langevin Dynamics

Human Evaluation

- No automated metric is perfect.
- The best way to evaluate generative models is by human assessment.
- It's important to assess various properties.

Human Evaluation

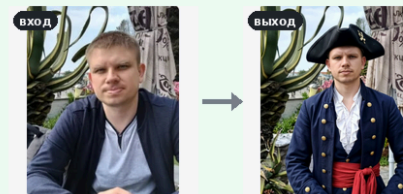
- No automated metric is perfect.
- The best way to evaluate generative models is by human assessment.
- It's important to assess various properties.

РЕЛЕВАНТНОСТЬ

«Выполнена ли инструкция?»

Сделал ли редактор именно то, что просили

✓ ХОРОШО



Запрос: «сделать пиратом». Образ пирата создан — то, о чём просили, выполнено.

✗ ПЛОХО



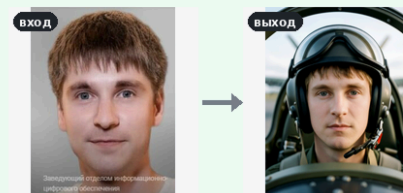
Запрос: эффект вращения на карусели. Картинка почти не изменилась — инструкция не выполнена.

СОХРАНЕНИЕ

«Сохранилось ли то, что менять не просили?»

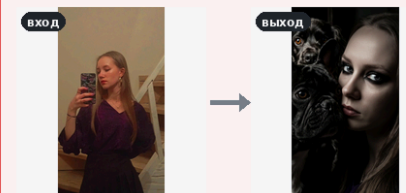
Фон, лицо и композиция остаются на месте

✓ ХОРОШО



Запрос: образ лётчика, «не меняя черты лица». Под шлемом — то же самое лицо.

✗ ПЛОХО



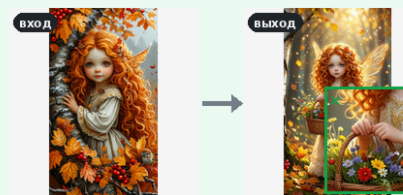
Запрос: «не меняй внешность, один в один», лишь приблизить лицо к собаке. Но на выходе — совершенно другой человек.

ДЕФЕКТЫ

«Не появилось ли артефактов?»

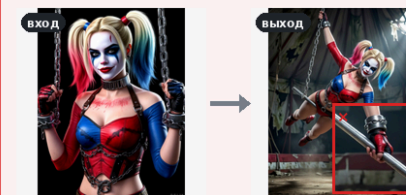
Лишние пальцы, сломанная геометрия и т. п.

✓ ХОРОШО



Кисти рук и предметы без искажений — артефактов нет.

✗ ПЛОХО



Кисть руки слилась в бесформенный ком (увеличено) — явный артефакт.

Outline

01 Generative Adversarial Networks (GAN)

02 Wasserstein Distance

03 Wasserstein GAN (WGAN)

04 Evaluation of Likelihood-Free Models

Frechet Inception Distance (FID) · Precision-Recall · CLIP Score · Human Eval

05 Langevin Dynamics

Energy-Based Models

Unnormalized Density

$$p_{\theta}(\mathbf{x}) = \frac{\hat{p}_{\theta}(\mathbf{x})}{Z_{\theta}}, \quad \text{where } Z_{\theta} = \int \hat{p}_{\theta}(\mathbf{x}) d\mathbf{x}$$

- $\hat{p}_{\theta}(\mathbf{x})$ can be any non-negative function.
- If we reparameterize as $\hat{p}_{\theta}(\mathbf{x}) = \exp(-f_{\theta}(\mathbf{x}))$, we eliminate the non-negativity constraint.

Energy-Based Models

Unnormalized Density

$$p_{\theta}(\mathbf{x}) = \frac{\hat{p}_{\theta}(\mathbf{x})}{Z_{\theta}}, \quad \text{where } Z_{\theta} = \int \hat{p}_{\theta}(\mathbf{x}) d\mathbf{x}$$

- $\hat{p}_{\theta}(\mathbf{x})$ can be any non-negative function.
- If we reparameterize as $\hat{p}_{\theta}(\mathbf{x}) = \exp(-f_{\theta}(\mathbf{x}))$, we eliminate the non-negativity constraint.

Log-Density Gradient

The gradient of the normalized log-density equals that of the unnormalized log-density:

$$\nabla_{\mathbf{x}} \log p_{\theta}(\mathbf{x}) = \nabla_{\mathbf{x}} \log \hat{p}_{\theta}(\mathbf{x}) - \nabla_{\mathbf{x}} \log Z_{\theta} = \nabla_{\mathbf{x}} \log \hat{p}_{\theta}(\mathbf{x})$$

Energy-Based Models

Unnormalized Density

$$p_{\theta}(\mathbf{x}) = \frac{\hat{p}_{\theta}(\mathbf{x})}{Z_{\theta}}, \quad \text{where } Z_{\theta} = \int \hat{p}_{\theta}(\mathbf{x}) d\mathbf{x}$$

- $\hat{p}_{\theta}(\mathbf{x})$ can be any non-negative function.
- If we reparameterize as $\hat{p}_{\theta}(\mathbf{x}) = \exp(-f_{\theta}(\mathbf{x}))$, we eliminate the non-negativity constraint.

Log-Density Gradient

The gradient of the normalized log-density equals that of the unnormalized log-density:

$$\nabla_{\mathbf{x}} \log p_{\theta}(\mathbf{x}) = \nabla_{\mathbf{x}} \log \hat{p}_{\theta}(\mathbf{x}) - \nabla_{\mathbf{x}} \log Z_{\theta} = \nabla_{\mathbf{x}} \log \hat{p}_{\theta}(\mathbf{x})$$

- Suppose we already have this density (normalized or not) $p_{\theta}(\mathbf{x})$.
- How can we sample from the model?

Langevin Dynamics

Theorem (Informal)

Let $\mathbf{x}_0$ be a random vector. Under mild regularity conditions, samples from the following dynamics will eventually follow $p_\theta(\mathbf{x})$ (for sufficiently small η and large l):

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}_l} \log p_\theta(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l, \quad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I}).$$

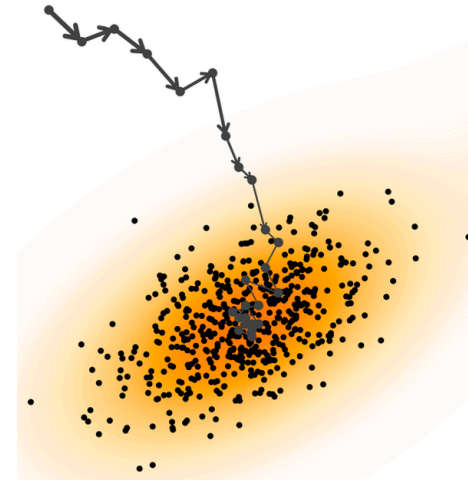
Langevin Dynamics

Theorem (Informal)

Let $\mathbf{x}_0$ be a random vector. Under mild regularity conditions, samples from the following dynamics will eventually follow $p_\theta(\mathbf{x})$ (for sufficiently small η and large l):

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}_l} \log p_\theta(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l, \quad \boldsymbol{\epsilon}_l \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$$

- What if $\boldsymbol{\epsilon}_l = \mathbf{0}$?
- The density $p_\theta(\mathbf{x})$ is the **stationary** distribution of the Markov chain.
- The gradient is taken with respect to $\mathbf{x}$, not $\boldsymbol{\theta}$.
- $\nabla_{\mathbf{x}} \log p_\theta(\mathbf{x})$ defines a vector field.



Langevin Dynamics: Why Add Noise?

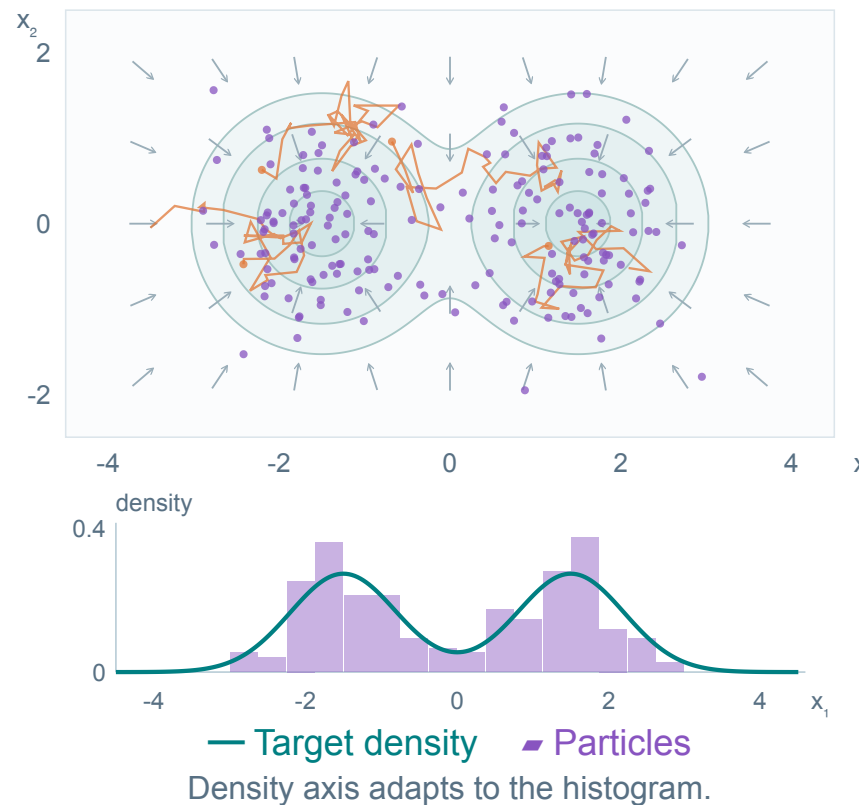
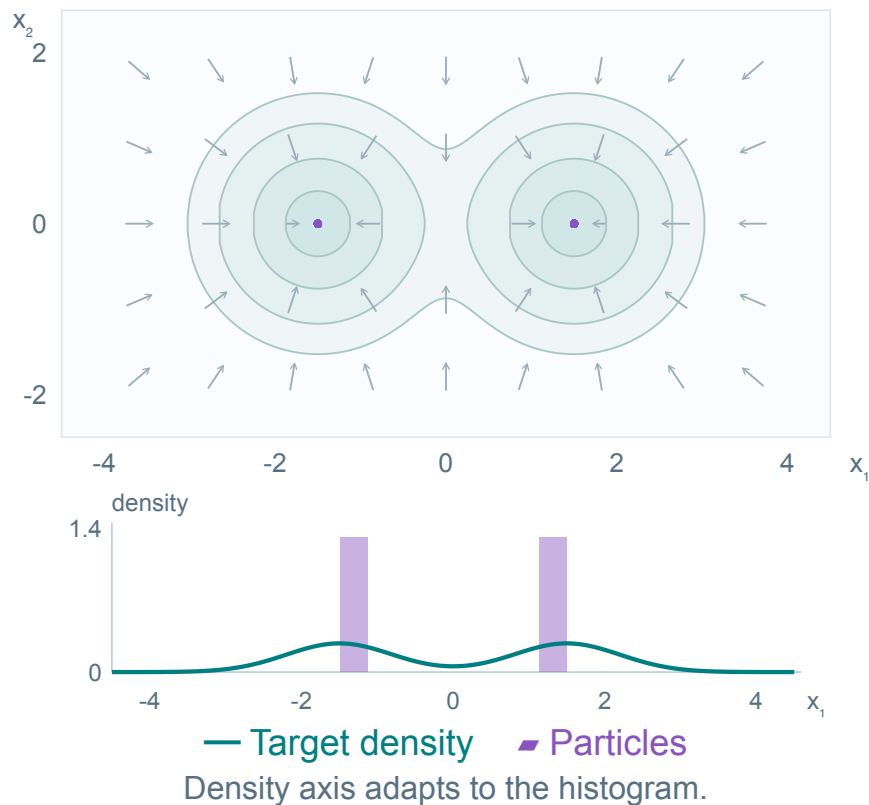
$$p_{\theta}(\mathbf{x}) = \frac{1}{2}\mathcal{N}(\boldsymbol{\mu}_{-}, 0.7^2\mathbf{I}) + \frac{1}{2}\mathcal{N}(\boldsymbol{\mu}_{+}, 0.7^2\mathbf{I}), \quad \boldsymbol{\mu}_{\pm} = (\pm 1.5, 0), \quad \eta = 0.04.$$

Noise off: approach the modes

Step 600

Noise on: explore the density

Step 600



Finite step size and finite simulation time give approximate sampling.

Summary

- GANs can suffer from mode collapse; KL and Jensen-Shannon measure mismatch differently.
- Perceptual and adversarial losses train autoencoders; learned priors model their latents.
- WGAN uses Kantorovich-Rubinstein duality to estimate a distance informative even for disjoint supports; weight clipping constrains the critic.
- FID compares Gaussian approximations; CMMD compares distributions of CLIP features using a kernel.
- Precision-recall measures the balance between sample quality and diversity.
- Image–text similarity, human preference, and compositional correctness require different evaluation signals.
- Langevin dynamics uses log-density gradients and Gaussian noise to sample from an unnormalized model.